

Unit-2

Sequence and Series

Upper Limit $\rightarrow \infty$

Greek letter 'Sigma' $\rightarrow \sum$

Lower Limit (where to start) $\rightarrow n=1$

n^2 $\leftarrow n^{\text{th}} \text{ term}$

$$\sum_{n=1}^{\infty} n^2$$



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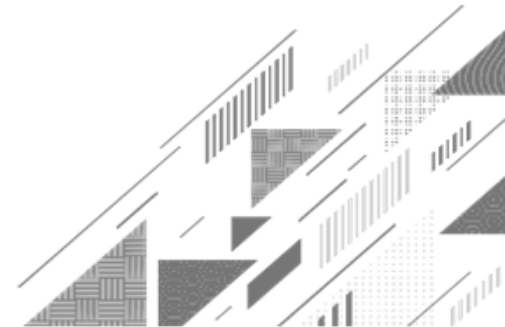
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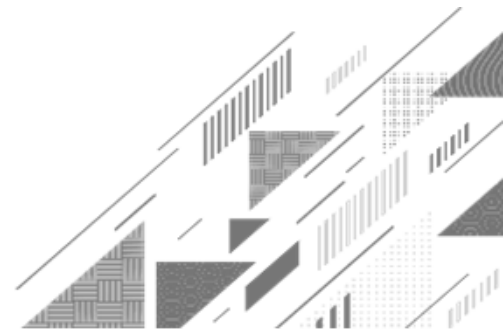


OUTLINE

- **Sequence**
- **Increasing and Decreasing Sequence**
- **The Sandwich Theorem OR Squeeze Theorem**
- **Convergent, Divergent and Oscillating Sequence**



Sequence



Sequence

Sequence

- ▶ A sequence is a set of numbers in a specific order.
- ▶ It is denoted by $\{a_n\}$ **OR** (a_n) **OR** $\{a_n\}_{n=1}^{\infty}$ **OR** $\{a_n\}_{n \geq 1}$.
- ▶ It is written as $\{a_n\} = a_1, a_2, a_3, \dots, a_n, \dots$

1^{st} term of $\{a_n\}$ 2^{nd} term of $\{a_n\}$ 3^{rd} term of $\{a_n\}$ n^{th} term of $\{a_n\}$

Sequence

Examples

$$(1). 2, 4, 6, 8, \dots = \{2n\}$$

$$(3). 1, 2, 3, 4, \dots = \{n\}_1^\infty$$

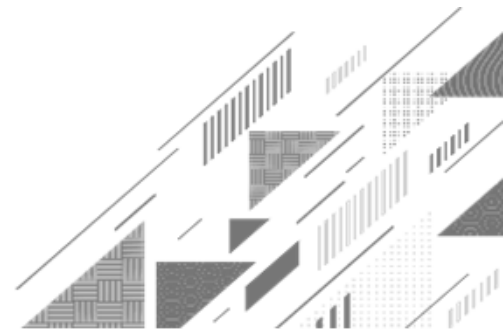
$$(2). 1, 3, 5, 7, \dots = (2n - 1)$$

$$(4). 2, 7, 12, 17, \dots = \{5n - 3\}_{n \geq 1}$$

$$(5). \left\{ \frac{n}{n+1} \right\}_{n=1}^\infty = \frac{1}{1+1}, \frac{2}{2+1}, \frac{3}{3+1}, \dots = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$$

$$(6). \left\{ \frac{1}{4^n} \right\}_{n \geq 3} = \frac{1}{4^3}, \frac{1}{4^4}, \frac{1}{4^5}, \dots$$

Increasing and Decreasing Sequence



Sequence

Increasing Sequence

$$a_n - a_{n+1} \leq 0, \forall n \in \mathbb{N}$$

► A sequence $\{a_n\}$ is said to be an increasing sequence if $a_n \leq a_{n+1}, \forall n \in \mathbb{N}$.

Examples (1). 1, 3, 5, 7, ...

$$a_1 = 1$$

$$a_2 = 3$$

$$a_3 = 5$$

$$a_4 = 7$$

⋮

Clearly, $a_1 \leq a_2 \leq a_3 \leq a_4 \dots$

Hence, given sequence is increasing.

Sequence

$$a_n - a_{n+1} \leq 0, \forall n \in \mathbb{N}$$

$$(2). \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5} \dots = \left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty} = \{a_n\}_{n=1}^{\infty}$$

$$\text{Here, } a_n = \frac{n}{n+1} \Rightarrow a_{n+1} = \frac{n+1}{n+2}$$

➔ Now, $a_n - a_{n+1}$

$$= \frac{n}{n+1} - \frac{n+1}{n+2}$$

$$= \frac{n(n+2) - (n+1)^2}{(n+1)(n+2)}$$

$$= \frac{\cancel{n^2} + \cancel{2n} - \cancel{n^2} - \cancel{2n} - 1}{(n+1)(n+2)}$$

$$= \frac{-1}{(n+1)(n+2)} \leq 0 \text{ as } n \in \mathbb{N}$$

$$\Rightarrow a_n - a_{n+1} \leq 0, \forall n \in \mathbb{N}.$$

Hence, given sequence is increasing.

Sequence

Decreasing Sequence

$$a_n - a_{n+1} \geq 0, \forall n \in \mathbb{N}$$

► A sequence $\{a_n\}$ is said to be a decreasing sequence if $a_n \geq a_{n+1}, \forall n \in \mathbb{N}$.

Examples (1). $-1, -3, -5, -7, \dots$

$$\text{Here, } a_1 = -1$$

$$a_2 = -3$$

$$a_3 = -5$$

$$a_4 = -7$$

$$\vdots$$

Clearly, $a_1 \geq a_2 \geq a_3 \geq a_4 \dots$

Hence, given sequence is decreasing.

Sequence

$$a_n - a_{n+1} \geq 0, \forall n \in \mathbb{N}$$

$$(2). \left\{ \frac{n}{n^2 + 1} \right\}_{n=1}^{\infty} = \frac{1}{2}, \frac{2}{5}, \frac{3}{10}, \frac{4}{17} \dots = \{a_n\}_{n=1}^{\infty}$$

$$\text{Here, } a_n = \frac{n}{n^2 + 1} \Rightarrow a_{n+1} = \frac{n+1}{(n+1)^2 + 2} = \frac{n+1}{n^2 + 2n + 3}$$

➡ Now, $a_n - a_{n+1}$

$$= \frac{n}{n^2 + 1} - \frac{n+1}{n^2 + 2n + 3}$$

$$= \frac{n(n+2) - (n+1)^2}{(n+1)(n+2)}$$

$$= \frac{n^3 + 2n^2 + 3n - n^3 - n^2 - n - 1}{(n+1)(n+2)}$$

$$= \frac{n^2 + 2n - 1}{(n+1)(n+2)} \geq 0 \text{ as } n \in \mathbb{N}$$

$$\Rightarrow a_n - a_{n+1} \geq 0, \forall n \in \mathbb{N}.$$

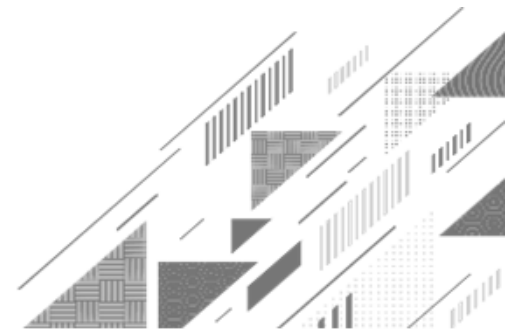
Hence, given sequence is decreasing.



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The Sandwich Theorem or Squeeze Theorem



The Sandwich Theorem OR Squeeze Theorem

► Let $\{a_n\}$, $\{b_n\}$ & $\{c_n\}$ be the sequence of real numbers.

If $a_n \leq b_n \leq c_n, \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Examples (1). $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = ?$

We know that $-1 \leq \sin n \leq 1, \forall n \in \mathbb{N}$

$$\Rightarrow -\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \forall n \in \mathbb{N}$$

Sandwich Theorem

$$\Rightarrow -\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \forall n \in \mathbb{N}$$

$$\text{Now, } \lim_{n \rightarrow \infty} -\frac{1}{n} = -\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \text{ \& } \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Hence, by **sandwich theorem**, $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$

The Sandwich Theorem OR Squeeze Theorem

(2). $\lim_{n \rightarrow \infty} \frac{1}{3^n} = 0$

► Let $\{a_n\}$, $\{b_n\}$ & $\{c_n\}$ be the sequence of real numbers.

We know that $-1 \leq (-1)^n \leq 1, \forall n \in \mathbb{N}$
If $a_n \leq b_n \leq c_n, \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Sandwich Theorem

We know that $-1 \leq (-1)^n \leq 1, \forall n \in \mathbb{N}$

$$\Rightarrow -\frac{1}{3^n} \leq \frac{(-1)^n}{3^n} \leq \frac{1}{3^n}, \forall n \in \mathbb{N}$$

$$\text{Now, } \lim_{n \rightarrow \infty} -\frac{1}{3^n} = -\lim_{n \rightarrow \infty} \frac{1}{3^n} = 0 \text{ \& } \lim_{n \rightarrow \infty} \frac{1}{3^n} = 0$$

Here,

The Sandwich Theorem OR Squeeze Theorem

$$n \rightarrow \infty \quad 3^n$$

► Let $\{a_n\}, \{b_n\}$ & $\{c_n\}$ be the sequence of real numbers.

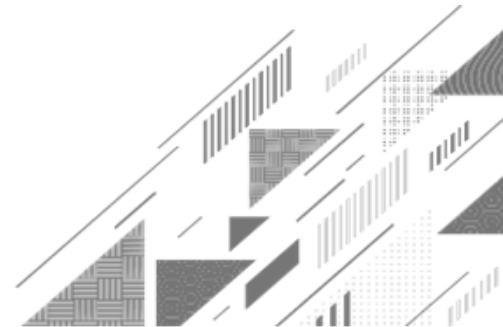
If $a_n \leq b_n \leq c_n, \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ then $\lim_{n \rightarrow \infty} b_n = L$.



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Convergent, Divergent and Oscillating Sequence



Sequence

Convergent Sequence

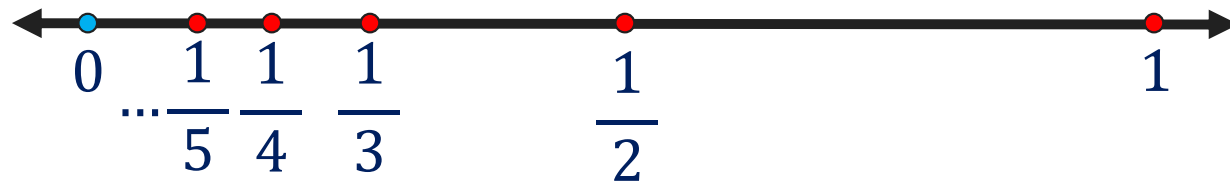
► A sequence $\{a_n\}$ is said to be convergent if $\lim_{n \rightarrow \infty} a_n = L$,
where a_n is n^{th} term of given sequence and 'L' is any finite number.

Example $\left\{ \frac{1}{n} \right\} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

$$L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$\Rightarrow L$ is finite.

Hence, given sequence is convergent.



Sequence

Divergent Sequence

► A sequence $\{a_n\}$ is said to be divergent if $\lim_{n \rightarrow \infty} a_n = \pm\infty$, where a_n is n^{th} term of given sequence.

Example $\{n^2\} = 1, 4, 9, 16, \dots$

$$L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} n^2 = \infty$$

$\Rightarrow L$ is infinite.

Hence, given sequence is divergent.



Sequence

Oscillating Sequence

▶ A sequence which is neither convergent nor divergent is said to be an oscillating sequence.

Example (1). $\{4 - (-1)^n\} = 3, 5, 3, 5, 3, 5, \dots$

$$\begin{aligned} L = \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \{4 - (-1)^n\} \\ &= \begin{cases} 4 - 1 = 3, & \text{if } n \text{ is even} \\ 4 - (-1) = 5, & \text{if } n \text{ is odd} \end{cases} \end{aligned}$$

Hence, given sequence is
Finitely oscillate between
3 & 5.

Sequence

Oscillating Sequence

▶ A sequence which is neither convergent nor divergent is said to be an oscillating sequence.

Example (2). $\{(-1)^n n^2\} = -1, 1, -4, 4, -9, 9, \dots$

The given sequence is neither diverges to $-\infty$ and to $+\infty$.

Therefore, the given sequence oscillate infinitely.

Method:2 → Convergence of Sequence

Example:1 Check the convergence of (1). $\left\{ \frac{n^2 + 1}{2n^2 - 1} \right\}$

Solution: Here, $a_n = \frac{n^2 + 1}{2n^2 - 1}$

→ Now, $\lim_{n \rightarrow \infty} \frac{n^2 + 1}{2n^2 - 1}$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^2} \left(1 + \frac{1}{n^2} \right)}{\cancel{n^2} \left(2 - \frac{1}{n^2} \right)}$$

$$= \frac{1 + 0}{2 - 0} \quad \left\{ \because \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \right\}$$

$$= \frac{1}{2}$$

→ Thus, the given sequence converges to 0.5

Method:2 → Convergence of Sequence

Example:1 Check the convergence of $(2). \{4^{n+1}\}$

Solution: Here, $a_n = 4^{n+1}$

→ Now, $\lim_{n \rightarrow \infty} 4^{n+1}$

$$= \lim_{n \rightarrow \infty} 4^n \cdot 4$$

$$= 4 \lim_{n \rightarrow \infty} 4^n = \infty$$

→ Thus, the given sequence is diverges to ∞ .

Method:2 ➡ Convergence of Sequence

Example:2 Define the convergence of a sequence (a_n) and verify whether the

sequence whose n^{th} term is $a_n = \left(\frac{n+1}{n-1} \right)^n$ converges or not?

Solution:

Convergent Sequence

► A sequence $\{a_n\}$ is said to be convergent if $\lim_{n \rightarrow \infty} a_n = L$,
where a_n is n^{th} term of given sequence and 'L' is any finite number.

Method:2 Example:2(Continue)

Example:2 Define the convergence of a sequence (a_n) and verify whether the

sequence whose n^{th} term is $a_n = \left(\frac{n+1}{n-1}\right)^n$ converges or not?

Solution: Here, $a_n = \left(\frac{n+1}{n-1}\right)^n$

 Now, $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1}\right)^n$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{n^n \left(1 + \frac{1}{n}\right)^n}{n^n \left(1 - \frac{1}{n}\right)^n} \\ &= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^n}{\left(1 - \frac{1}{n}\right)^n} \end{aligned}$$

Method:2 $\rightsquigarrow$ Example:2(Continue)

$$= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^n}{\left(1 - \frac{1}{n}\right)^n}$$

$$= \frac{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n}{\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n}$$

➡ We know that, $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

$$\& \lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n = e^{-x}$$

$$= \frac{e^1}{e^{-1}}$$

$$= e^2$$

Mathematics-I
GTU#(3110014)

**Thank
You**

Upper Limit $\rightarrow \infty$

Greek letter 'Sigma' $\rightarrow \sum$

Lower Limit (where to start) $\rightarrow n=1$

n^2 $\leftarrow n^{\text{th}} \text{ term}$



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